POWER BI ANALYTICS SOLUTION

E-Commerce Business Intelligence

Case Study & Implementation Blueprint

Transforming Fragmented Data into Actionable Revenue Intelligence

Executive Summary

A high-growth e-commerce company operating across multiple cities, product categories, and customer segments faced a critical visibility gap: sales, customer behaviour, pricing, and promotional data lived in separate silos with no unified analytical layer. Leadership, marketing, and operations teams were making decisions in the dark.

This document presents a complete Power BI analytics solution that transforms fragmented raw data into a single, trusted intelligence platform. The solution is organized around four business outcomes:

Revenue & Pricing Intelligence	Customer Behaviour Analytics	Promotional Effectiveness	Operational Performance

IMPACT	By unifying sales events, promotions, and supporting reference data in Power BI, the company expects to recover up to 18–25% in missed revenue opportunities, reduce customer acquisition costs through smarter campaign targeting, and accelerate time-to-insight from days to minutes.

1. Business Context & The Problem Statement

1.1 Where the Business Stands Today

The company has scaled rapidly — expanding SKUs, entering new city tiers, and growing its digital channels. But this growth has created an analytics debt. The data exists; the ability to use it does not. Three major pain points define the current state:

Pain Point	Business Impact	Root Cause
No single view of performance	Leadership relies on weekly Excel exports; decisions lag reality by 7–10 days	Data scattered across sales, promo, and logistics systems
Pricing has no analytical guardrails	Margin leakage on discounted SKUs goes undetected until month-end close	No real-time pricing vs. sales correlation model
Promotions are run on instinct, not data	Marketing spends budget on campaigns with no measurable engagement lift	Promotional events not linked back to actual purchase conversions
Customer journey is invisible	Unable to distinguish high-value customers from one-time buyers at campaign time	Event data (views, carts, purchases) not integrated or modelled

1.2 What the Business Needs to Solve

The analytics team has been mandated to deliver a Power BI solution that answers five core business questions — not as one-off reports, but as live, filterable intelligence that any stakeholder can use independently:

- How is pricing behaving across brands, categories, time, and channel — and where is it hurting revenue?
- Who are our customers, when do they engage, and what drives them to buy vs. browse?
- Are our promotions working — and for which segments, categories, and periods do they create the most lift?
- Where is operational performance strong, and where are city tiers or channels underperforming?
- What signals in the data should trigger an immediate business action?

2. Data Landscape & Integration Architecture

2.1 Source Systems Overview

The solution integrates data from five distinct source layers. Understanding what each contains — and its quality limitations — is essential before modelling begins.

Source	Key Contents	Update Frequency	Data Quality Risk
Sales / Events Data	Customer interactions: product views, cart additions, purchases, channel (App/Browser), timestamps	Daily	Duplicate events, null product IDs
Promotion Data	Promo codes, discount %, validity windows, eligible categories	Per campaign	Overlapping promo periods, unmapped SKUs
Product Mapping	SKU-to-category-to-brand hierarchy	Monthly	Discontinued SKUs without flags
City / Pin Code Tiers	Geographic classification (Tier 1/2/3), state mapping	Quarterly	Pin codes not mapped to tiers
Pricing Reference	Listed price, MRP, discount bands by SKU and channel	Weekly	Price changes mid-period without versioning

2.2 Data Integration Strategy in Power BI

Rather than loading all tables into a flat model, the solution implements a star schema with a clearly defined fact-dimension architecture. This ensures query performance, filter propagation, and DAX simplicity across all dashboards.

Table Type	Tables	Role in Model
Fact Tables	Fact_SalesEvents, Fact_Promotions	All transactional and measurable event data; connected via keys to dimensions
Dimension Tables	Dim_Product, Dim_Customer, Dim_Geography, Dim_Channel, Dim_Date	Descriptive context enabling filtering and grouping
Bridge Tables	Bridge_PromoProduct	Resolves many-to-many between promotions and eligible SKUs

2.3 Key Derived Columns & Enrichment

Raw data alone cannot answer business questions. The following derived columns are created during Power Query transformation to enrich the analytical model:

- Day Type: Weekday vs. Weekend flag on every transaction date — enables day-of-week demand patterns
- Session Type: App vs. Browser derived from channel event metadata — critical for channel mix analysis
- Revenue Realized vs. Potential Revenue: Actual purchase value vs. MRP × units — quantifies discount leakage
- Net Units: Purchased units minus returned units — cleans up inflated sales figures
- Funnel Stage: Categorises events into View → Cart → Purchase sequence — enables drop-off analysis
- Promo Active Flag: Boolean indicating whether a promotion was live on the transaction date for that SKU
- City Tier: Mapped from pin code via lookup — enables Tier 1/2/3 performance comparison

3. Power BI Dashboard Solution

The solution delivers four business-facing dashboards, each designed for a specific stakeholder audience. Every dashboard is filterable by date range, category, brand, channel, and geography from a shared filter pane.

Dashboard 1: Executive Revenue Command Centre

Audience: CEO, CFO, Business Heads

This is the company's single pane of glass for revenue performance. It answers the first question leadership asks every morning: How are we tracking?

BUSINESS GOAL Provide leadership with real-time visibility into revenue, conversion, and growth — broken down by month, geography, and channel — so course-corrections happen in days, not weeks.

Key Metrics Tracked

KPI	Definition	Why It Matters
Total Revenue	Sum of purchase value (net of returns)	Primary performance indicator
Potential Revenue	Units viewed × MRP; what could have been earned	Reveals opportunity cost of unsold inventory
Revenue Capture Rate	Revenue ÷ Potential Revenue × 100	Measures how effectively demand converts to sales
Month-over-Month Growth %	Revenue this month vs. last month	Early warning indicator for demand shifts
Active Products & Categories	Count of SKUs and categories with at least 1 purchase	Monitors breadth of revenue contribution

Visualisations in This Dashboard

- Revenue Trend Line: Monthly revenue plotted against potential revenue — gap signals pricing or conversion problems
- State-Level Heat Map: Revenue intensity by state — highlights geographic concentration risk
- Channel Mix Bar: App vs. Browser revenue share by month — tracks digital channel evolution
- Top 10 Categories Card: Sorted by revenue contribution — reveals portfolio performance
- Revenue vs. Target Gauge: Actual vs. monthly target — drives operational accountability

INSIGHT EXAMPLE If the Revenue Capture Rate drops below 60% in a given month while Potential Revenue is rising, it signals demand exists but pricing or promotions are failing to convert it — a direct trigger for the pricing and promotions team to investigate.

Dashboard 2: Customer Behaviour & Engagement Analytics

Audience: Marketing, CRM, Digital Teams

Understanding when customers engage, what they look at, and why they do or do not purchase is the foundation of effective marketing. This dashboard maps the complete customer journey from first touch to conversion.

BUSINESS GOAL

Identify the highest-value customer touchpoints, time windows, and channels so that marketing budget is allocated to moments of maximum impact — not broadcast evenly across all sessions.

Customer Journey Funnel Analysis

Every customer interaction is categorised into one of three funnel stages. The drop-off between stages is the most actionable metric in this dashboard:

Funnel Stage	Event Type	Drop-off Insight
Awareness (View)	Product page view events	High views + low carts = product appeal issue or pricing mismatch
Consideration (Cart)	Add-to-cart events	High carts + low purchases = checkout friction or price sensitivity
Conversion (Purchase)	Completed transaction events	Conversion rate by segment reveals who is buying and who is not

Traffic Pattern Analysis

- Day-of-Week Heatmap: Sessions and purchases by weekday — reveals peak demand windows (e.g., Sunday spikes in App traffic)
- Hourly Engagement Chart: Transactions bucketed by hour of day — identifies prime-time windows for promotions
- App vs. Browser Conversion Comparison: Conversion rate split by channel — typically App users convert at 1.4–2× the rate of Browser users
- Brand Preference by Category: Which brands drive engagement within each category — informs assortment and placement strategy
- Search Behaviour Patterns: Category searches by brand and brand searches within categories — surfaces unmet demand signals

INSIGHT EXAMPLE

If Browser traffic peaks Tuesday–Thursday but App traffic peaks Saturday–Sunday with 40% higher conversion, the marketing team should shift weekend push notifications to App-exclusive promotions and redirect mid-week email spend toward categories with high Browser cart rates.

Dashboard 3: Pricing Intelligence & Margin Optimisation

Audience: Category Managers, Pricing Team, Finance

Pricing is the single largest lever in e-commerce profitability, yet most companies operate without real-time visibility into how pricing decisions play out across brands, channels, and time. This dashboard closes that gap.

BUSINESS GOAL	Surface where discounting is destroying margin without driving proportional volume, and where pricing has headroom to increase without hurting conversion.

Pricing Analytics Framework

The dashboard operates across three analytical layers:

Layer 1 — Pricing Structure Visibility

- Listed Price vs. MRP by Brand/Category: Identifies categories with chronic over-discounting
- Discount Band Distribution: % of SKUs in 0–10%, 10–25%, 25–40%, and 40%+ discount ranges
- Price Elasticity Scatter Plot: Discount depth on X-axis, units sold on Y-axis — identifies elastic vs. inelastic SKUs

Layer 2 — Channel Pricing Comparison

- App vs. Browser Price Differential Matrix: Where different prices exist for the same SKU across channels — flags potential customer trust risks
- Discount by Channel Over Time: Trends in discount depth per channel — reveals if App users are being trained to expect higher discounts

Layer 3 — Revenue Impact of Pricing Decisions

- Revenue Lost to Discounting: Difference between full-price potential revenue and actual revenue, by category
- Margin Recovery Opportunity: SKUs where reducing discount by 5–10% would have minimal conversion impact based on elasticity data

INSIGHT EXAMPLE	If Brand X in Category Y shows a discount depth of 38% but the elasticity data shows no meaningful volume increase compared to 25% discount on the same SKU, Category Managers have a data-backed case to reduce the discount floor — recovering margin without volume loss.

Dashboard 4: Promotional Effectiveness Tracker

Audience: Marketing, Trade Marketing, Growth Teams

Promotions without measurement are costs, not investments. This dashboard transforms promotional spend from a faith-based activity into a measurable revenue driver — or exposes where it is not working.

BUSINESS GOAL

Measure the true incremental lift each promotion generates in sales, traffic, and customer engagement — and identify which promotional structures deliver the best return across segments, categories, and time periods.

Promotional Measurement Methodology

Measurement	Calculation	Business Use
Promo Lift — Traffic	Sessions during promo vs. equivalent non-promo period	Did the promotion actually drive more visitors?
Promo Lift — Conversion	Purchase rate during vs. outside promo window	Did more visitors actually buy?
Promo Lift — Revenue	Actual revenue during promo vs. baseline forecast	Did total revenue increase, or did discounts just cannibalise full-price sales?
Discount Efficiency Ratio	Revenue lift ÷ Discount given away	How much incremental revenue was generated per rupee of discount
Customer Acquisition via Promo	New-to-brand buyers during promo period	Is the promotion growing the customer base or rewarding existing buyers?

Promotional Segmentation Insights

- Category-Level Promo Response: Which categories see the strongest lift — typically electronics and fashion outperform staples
- Brand-Level Promo Sensitivity: Some brands' customers are highly promo-driven; others convert equally at full price
- City Tier Promo Impact: Promotions often over-index in Tier 2 and Tier 3 cities where price sensitivity is higher
- Weekday vs. Weekend Promo Timing: Flash promotions on weekends combined with high App usage create amplified lift windows

INSIGHT EXAMPLE

A 3-day flash promotion on a fashion brand runs across all city tiers. The dashboard reveals: Tier 1 cities showed only 8% revenue lift (customers already buying), while Tier 2 cities showed 34% lift. Future editions of this promotion should be targeted exclusively at Tier 2 and Tier 3 with App-push activation, reducing discount cost while maintaining lift.

4. Implementation Roadmap

The solution is delivered in three phases over eight weeks. Each phase produces working outputs that stakeholders can begin using immediately, while subsequent phases add depth and sophistication.

Phase	Deliverables	Stakeholder Benefit
Phase 1	Data connections established, Power Query cleaning scripts, star schema model, Date/Geography dimensions built	Data team has a clean, reliable foundation; no more manual Excel consolidation
Phase 2	Executive Revenue dashboard live, Customer Behaviour dashboard live, core DAX measures published to shared dataset	Leadership gets daily revenue visibility; marketing gets funnel and traffic analysis
Phase 3	Pricing Intelligence dashboard, Promotional Effectiveness tracker, row-level security by role, scheduled refresh configured	Category managers and growth teams get self-serve pricing and promo analytics

4.1 Technical Configuration

- Refresh Schedule: Incremental daily refresh for sales events (large table); full refresh weekly for pricing and product reference tables
- Row-Level Security: Operations team sees only their assigned region; marketing sees all channels; leadership has unrestricted access
- Performance Optimisation: Aggregation tables built for Executive dashboard to ensure sub-3-second load on full data volumes
- Power BI Service Deployment: Published to a dedicated workspace with app-based distribution; alerts configured on Revenue Capture Rate and Day-over-Day decline thresholds

5. Strategic Opportunities & Next Steps

5.1 Derived Analytics That Add Immediate Value

Beyond the four dashboards, the data model enables several high-value analytical extensions that can be activated without new data sources:

- Customer Segmentation: RFM (Recency, Frequency, Monetary) scoring using purchase event history — enables targeted retention and win-back campaigns
- Abandoned Cart Analysis: Users who added to cart but did not purchase within 48 hours — the highest-ROI remarketing audience
- Price Elasticity Modelling: Regression of discount depth vs. units sold per SKU — enables data-driven pricing floor recommendations
- Cohort Retention Analysis: Tracks whether customers acquired during a promotion return at full price — reveals true promotional customer quality

5.2 Additional Data Sources to Unlock Deeper Intelligence

Data Source	What It Enables	Priority
Customer Demographics (Age, Gender)	Personalised category recommendations, targeted campaign segmentation	High
Logistics & Delivery Data	Correlate delivery speed with repeat purchase rates; identify fulfilment-driven churn	High
Returns & Refund Data	Net revenue accuracy; identify quality issues by brand or category driving high returns	High
Competitor Pricing Feeds	Dynamic pricing context; identify where the company is uncompetitively priced	Medium
Customer Ratings & Reviews	Sentiment analysis by brand/category; early warning on quality degradation	Medium
Marketing Spend by Channel	Attribution modelling; connect marketing investment directly to revenue outcomes	High

5.3 Dashboard Governance & Adoption

- Stakeholder Onboarding: Structured walkthrough sessions with each audience (leadership, marketing, operations) using realistic scenarios specific to their decisions
- Feedback Loop: Monthly dashboard review sessions where business users surface gaps; ensures the solution evolves with business needs
- Data Dictionary: Published glossary of every KPI definition — eliminates ambiguity in leadership reviews about which number is the source of truth
- Alert & Subscription Setup: Proactive alerts configured for threshold breaches (e.g., Revenue Capture Rate below 55%), so insights reach stakeholders without requiring dashboard visits

6. Expected Business Outcomes

The success of this solution is measured not by dashboards delivered, but by business decisions improved. The following outcomes are tracked at the 90-day mark post-deployment:

Outcome	Measurement	Target
Faster revenue reporting	Time from month-end to leadership review	From 5–7 days to same-day
Pricing discipline improvement	Average discount depth across catalogue	Reduce by 4–6 percentage points
Promotional ROI improvement	Discount Efficiency Ratio	Improve by 20% in first two promotion cycles post-deployment
Marketing efficiency	Cost per acquired customer via promotions	Reduce 15–20% through targeted promo deployment
Revenue Capture Rate	Actual revenue ÷ potential revenue	Increase from baseline to 70%+ within 6 months
Self-serve analytics adoption	% of business decisions made without analyst support	60%+ of regular reporting queries handled by dashboard users independently

CLOSING NOTE

This Power BI solution is not a reporting project. It is a decision infrastructure project. When business users can answer their own questions, trust the numbers they see, and act on alerts before problems escalate — the analytics function shifts from cost centre to competitive advantage. That is the standard this solution is built to meet.

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